

Equations and Inequalities Problem Solving

Lesson 8-7

Name:

Date:

Class:



TODAY'S OBJECTIVES

Content Objective: I can model and solve real-world problems using equations and inequalities.

Language Objective: I can explain my reasoning using the words model, equation, inequality, and reasonableness.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Model

Equation

Inequality

Reasonableness

Variable

Constraint



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A drawing, equation, or diagram that represents a situation is a

2

A math sentence with an equal sign is an .

3

A math sentence that compares values with $<$, $>$, $\leq$, or $\geq$ is an

4

Checking whether an answer makes sense for the problem is checking its

- 5 A letter that stands for the unknown amount in a problem is a

.

- 6 A limit that tells which values are allowed in a problem is a

.

Write About the Math

How much more can the detective spend? How much does each flashlight cost? When do you use an equation versus an inequality?

¿Cuánto más puede gastar la detective? ¿Cuánto cuesta cada linterna? ¿Cuándo usas una ecuación y cuándo una desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Model**
Modelar

☐ **Equation**
Ecuación

☐ **Inequality**
Desigualdad

☐ **Reasonableness**
Razonabilidad

☐ **Variable**
Variable

Model: An exact total points to an equation; a limit or range points to an inequality. — Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can spend at most \$__ more because $r \leq 500 - __ = __$. Each flashlight costs \$__ because $f = 60 \div __ = __$.**
- **My answer is __ because __.**
Mi respuesta es __ porque __.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is __.**
Mi afirmación es __.
- **My evidence is __, so __.**
Mi evidencia es __, entonces __.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.